

Land and Infinite Debt Rollover¹

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Question

- Can government roll over debt forever, without raising taxes or cutting spending?
- Growing literature on infinite debt rollover with low interest rates (Blanchard, 2019; Mian et al., 2022; Ball and Mankiw, 2023; Kocherlakota, 2023; Abel and Panageas, 2025)
 - Benchmark: Diamond (1965) OLG with $R < G$ (dynamic inefficiency)
 - Economies typically **without productive land**
- Older literature: land precludes rollover (McCallum, 1987; Homburg, 1991, 2014)
- **Does productive land preclude infinite debt rollover?**

Classical argument

- On balanced growth path, land price grows at output growth rate G
- No arbitrage:

$$R = \frac{P_{t+1} + r_{t+1}}{P_t} = G \left(1 + \frac{r_{t+1}}{P_{t+1}} \right) > G$$

- Debt satisfying $D_{t+1} = RD_t$ grows faster than economy
*“the rate of return on land must exceed the growth rate.
[. . .] Oversaving or inefficient accumulation of excess capital in the sense of Phelps is shown to be impossible.”*

Nichols (1970, p. 339)

But land may become inessential

- Balanced growth with fixed land and growing labor requires (essentially) Cobb–Douglas production
- CES production with elasticity of substitution $\sigma > 1$:

$$\text{land share } s_t^X = \frac{1 - \alpha}{\alpha H_t^{1-1/\sigma} + 1 - \alpha} \rightarrow 0$$

- Land remains **technologically productive** but becomes **economically inessential** (Rhee, 1991)
- Rent need not grow with output, so classical argument breaks down
- Mere presence of land does not settle rollover question

Main results

1. **Land precludes rollover** if land share bounded away from zero
 - Every equilibrium fundamental and Pareto efficient
 - No preference or demographic restrictions
2. **Land enables rollover** if land share vanishes and autarky interest rate $<$ rent growth
 - Every equilibrium contains land bubble
 - Government debt can replace part of bubble and be rolled over forever
 - Every equilibrium Pareto efficient (under regularity condition)
3. **Iowa farmland**
 - Declining cash yield
 - Common component of rent-adjusted land prices rising since late 1990s

Related literature

- **Land and dynamic efficiency:** Nichols (1970), McCallum (1987), Homburg (1991), Rhee (1991), Geanakoplos (2005)
- **Debt rollover:** Blanchard (2019), Mian et al. (2022), Ball and Mankiw (2023), Kocherlakota (2023), Abel and Panageas (2025)
- **Rational bubbles and debt:** Tirole (1985), O'Connell and Zeldes (1988), Santos and Woodford (1997), Kocherlakota (2013)
- **Bubbles attached to dividend-paying assets:** Hirano and Toda (2025a), Hirano and Toda (2025b), Pham and Toda (2026)
- **Farmland valuation and bubble tests:** Burt (1986), Falk (1991), Just and Miranowski (1993), Giglio et al. (2016), Tolhurst (2021)

Environment

- Deterministic economy, $t = 0, 1, \dots$, single good
- Agents $i \in I$ with arbitrary finite or infinite lifetimes
 - locally nonsatiated preferences, no further restrictions
 - inelastic labor supply h_{it}
- Land: productive, non-reproducible, unit supply
- Production $Y_t = F_t(H_t, 1)$, constant returns

$$w_t = F_{t,H}(H_t, 1), \quad r_t = F_{t,X}(H_t, 1), \quad Y_t = w_t H_t + r_t$$

- Equivalent to endowment economy where land pays dividend r_t

Equilibrium and land bubbles

- Agents trade land sequentially; no arbitrage gives

$$R_t = \frac{P_{t+1} + r_{t+1}}{P_t}$$

- Fundamental value: present value of rents

$$V_t = \frac{1}{q_t} \sum_{s=t+1}^{\infty} q_s r_s, \quad q_t = \frac{1}{R_0 \cdots R_{t-1}}$$

- Bubble $B_t := P_t - V_t \geq 0$ grows at interest rate:

$$B_{t+1} = R_t B_t$$

Government debt

- Government issues $D_0 \geq 0$, transfers proceeds to initial landowners
- Zero primary balance thereafter:

$$D_{t+1} = R_t D_t$$

Lemma (Debt–bubble equivalence)

Equilibrium with positive infinite debt rollover exists if and only if economy without debt has bubbly equilibrium. Every issuance $D_0 \in (0, B_0]$ is feasible, with same allocation and interest rates.

- Bonds and land earn same return $\implies$ perfect substitutes
- Adapts Tirole (1985), O'Connell and Zeldes (1988), Kocherlakota (2013)

Fiscal interpretation

- Lemma establishes feasibility, not welfare gain
 - debt only replaces part of preexisting private bubble
 - issuance cannot exceed bubble B_0
 - spending proceeds would change allocation
- Feasibility of rollover need not reveal dynamic inefficiency
- Rollover question reduces to:

Does land price contain bubble?

Land-share criterion

Land share $s_t^X := r_t/Y_t$

Theorem

If $\liminf_{t \rightarrow \infty} s_t^X > 0$, every equilibrium is fundamental and Pareto efficient. Consequently, positive infinite debt rollover is impossible.

- Proof:
 1. $r_t \geq \delta Y_t$ and $\sum_t q_t r_t = V_0 \leq P_0 < \infty$ imply $\sum_t q_t Y_t < \infty$
 2. Finite-valued resources: first welfare theorem and no bubble (Santos and Woodford, 1997)
 3. Debt–bubble equivalence rules out rollover
- No restrictions on demographics or preferences, no convergence

► Detailed proof

Cobb–Douglas and classical argument

- Cobb–Douglas $F_t(H, X) = A_t H^\alpha X^{1-\alpha}$: $r_t = (1 - \alpha) Y_t$
- Corollary: every equilibrium fundamental and efficient, no rollover
- McCallum (1987) assumes convergence to steady state
 - comes close to requiring Cobb–Douglas (McCallum, 1987, Footnote 18)
- Homburg (1991), Rhee (1991): land share bounded away from zero in two-period OLG
- Operative restriction: **land share**, not balanced growth

Vanishing land share

Corollary

If positive infinite debt rollover is possible, then $\liminf_{t \rightarrow \infty} s_t^X = 0$.

- CES production with $H_t \rightarrow \infty$:

$$F_t(H, X) = A_t \left(\alpha H^{1-1/\sigma} + (1 - \alpha) X^{1-1/\sigma} \right)^{\frac{1}{1-1/\sigma}}$$

- $s_t^X \rightarrow 1$ if $\sigma < 1$, $s_t^X = 1 - \alpha$ if $\sigma = 1$, $s_t^X \rightarrow 0$ if $\sigma > 1$
- $\sigma > 1$ necessary but not sufficient for rollover
 - also need demand for saving

Two-period OLG

- Cohort of mass N_t born at t ; utility $U_t(c_t^y, c_{t+1}^o)$
- Labor endowments (h_t^y, h_t^o) , induced incomes $(e_t^y, e_t^o) = (w_t h_t^y, w_t h_t^o)$
- Budget constraints:

$$\begin{aligned}c_t^y + P_t x_t &= e_t^y, \\c_{t+1}^o &= e_{t+1}^o + (P_{t+1} + r_{t+1})x_t\end{aligned}$$

- Market clearing: $N_t x_t = 1$
- Aggregate young and old income: $a_t := N_t e_t^y$, $b_t := N_{t-1} e_t^o$

Three growth rates

- G : asymptotic growth rate of aggregate income a_t
- $G_r := \limsup_{t \rightarrow \infty} r_t^{1/t}$: asymptotic growth rate of rent
- R^a : **autarky interest rate**
 - limiting marginal rate of substitution if asset trade became negligible
 - counterfactual, not equilibrium interest rate
- Equilibrium **asymptotically bubbly** if bubbly and

$$\liminf_{t \rightarrow \infty} P_t/a_t > 0$$

Bubble necessity

Proposition

Under regularity conditions, if

$$R^a < G_r < G,$$

then equilibrium exists and every equilibrium is asymptotically bubbly.

- Suppose $P_t/a_t \rightarrow 0$ (along subsequence)
- Then allocation approaches autarky, interest rate approaches R^a
- $R^a < G_r$: present value of rents infinite, contradiction
- $G_r < G$: rent negligible relative to economy, but land price not
- Combines Theorems 1–2 of Hirano and Toda ([2025a](#))

Pareto efficiency and debt rollover

Theorem

*Under additional regularity conditions and $R^a < G_r < G$, every equilibrium is asymptotically bubbly and **Pareto efficient**.*

Government can roll over debt $D_0 \in (0, B_0]$ forever without changing allocation.

- Bubble grows at interest rate yet remains comparable to income
- Hence $R = G$ on average, present value of income infinite
- Economy lies on boundary of Cass criterion
- **Neither bubble nor rollover requires dynamic inefficiency**

Proof idea: Cass criterion

- Detrend by $\Lambda_t := N_t \lambda_t$ (population $\times$ efficiency units)
 - constant-population OLG economy
 - detrended rent $r_t/\Lambda_t \rightarrow 0$ because $G_r < G$: bounded resources
 - detrended prices $\hat{q}_t = q_t \Lambda_t / \Lambda_0$, land price $p_t = P_t / \Lambda_t$
- Cass criterion (Cass, 1972): with uniform curvature, equilibrium efficient if

$$\sum_{t=0}^{\infty} \frac{1}{\hat{q}_t} = \infty$$

- Bubble: $q_t P_t \rightarrow B_0 > 0$ and $\liminf p_t > 0$, so

$$\frac{1}{\hat{q}_t} = \frac{\Lambda_0 p_t}{q_t P_t} \quad \text{bounded away from zero}$$

- Contrast with Theorem 1: here $\sum_t q_t Y_t = \infty$, yet efficient

Comparison with pure bubbles

- Pure bubble (fiat money) (Tirole, 1985)
 - bubbly equilibrium may be efficient
 - but inefficient bubbleless equilibrium remains
- Land pays rents
 - $R^a < G_r < G$ eliminates asymptotically bubbleless branch
 - every equilibrium bubbly and efficient
- Bubble necessity for dividend-paying asset is subtle
 - Tirole's Proposition 1(c) incorrect as stated (Pham and Toda, 2026)
 - Rhee (1991) relies on Tirole's argument

CES production

- Population growth G_N , labor-endowment growth G_H , productivity growth G_A , $G_N G_H > 1$
- Preferences stationary in efficiency units, $\mathcal{R} := U_1/U_2$
- Growth rates:

$$G = G_A G_N G_H, \quad G_r = G_A (G_N G_H)^{1/\sigma}, \quad R^a = G_A G_H \mathcal{R}(e^y, e^o)$$

- $\sigma > 1$ implies $G_r < G$; $R^a < G_r$ if and only if

$$\mathcal{R}(e^y, e^o) < G_N^{1/\sigma} G_H^{1/\sigma-1}$$

- CRRA: $(1/\beta)(h^o/h^y)^\gamma < G_N^{1/\sigma} G_H^{1/\sigma-1}$; holds automatically if $h^o = 0$

Example: linear production

- Rent $r_t = rG^{t/\sigma}$, production $F_t(H, X) = H + r_tX$, CRRA utility
- Construct endowments supporting given land-price path
- Bubbly: $P_t = pG^t$
 - asymptotically bubbly and Pareto efficient
 - rollover of any $D_0 \in (0, B_0]$ feasible
- Fundamental: $P_t = pG^{t/\sigma}$ (with $e^o > 0$)
 - interest rate $R = G^{1/\sigma}(1 + r/p)$
 - efficient if and only if $R \geq G$
- Efficiency and rollover are separate questions

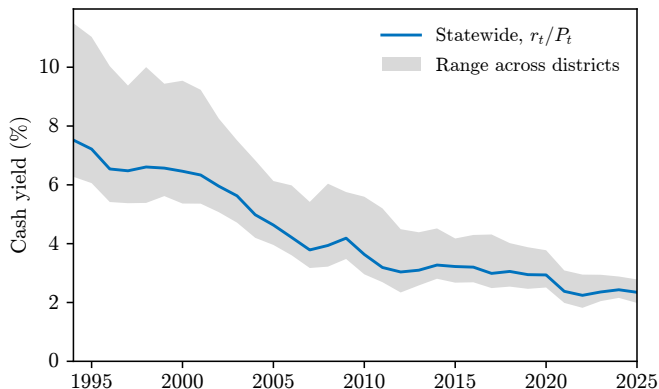
Why farmland

- Housing rent mixes returns to structure and underlying land
 - stationary housing rent–price ratio can coexist with land bubble (Toda, 2025)
- Agricultural rents and values per acre: cleaner counterpart to land in model
- Agriculture: land-intensive sector with declining output share

Data: Iowa State University (ISU) surveys

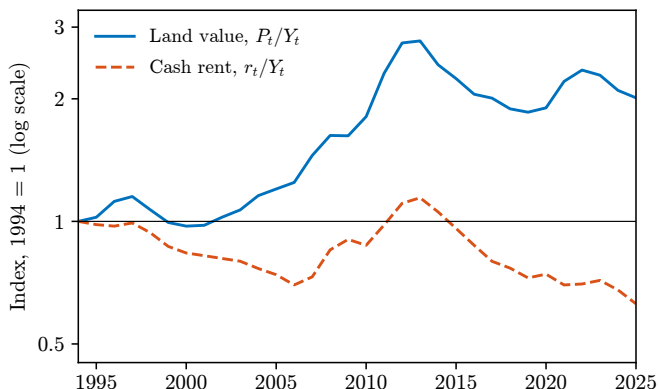
- Land values: state, 9 crop reporting districts, 99 counties, 1950–2025
- Cash rents (corn and soybean land): counties, 1994–2025
- Real values deflated by CPI

Cash yield



- Statewide cash yield: 7.5% (1994) → 2.3% (2025)
- Decline in every district

Rent and land value relative to GDP



- Rent: -37% relative to US GDP, consistent with $G_r < G$
- Land value: doubled relative to US GDP

Not yet evidence of bubble

- Lower interest rates or higher expected rent growth also raise price–rent ratio
- Need restrictions more specific to bubble
- Idea: use cross section of 9 districts
 1. rents measure relative fundamental values
 2. no arbitrage implies district bubbles share common growth factor

Common factor in district rents

- Factor structure:

$$r_{it} = \psi_i r_t \exp(\epsilon_{it})$$

- If exact, fundamental value $V_{it} = \psi_i V_t$
- Estimates:
 - common factor explains 97% of within-district variation
 - standard deviation of ϵ_{it} : 0.031
 - loadings $\log \hat{\psi}_i$ from 1994–2009 and 2010–2025: correlation 0.91
- Approximate, not exact: district slopes on $\log r_t$ range 0.81–1.18

► Slopes

Common bubble factor

- District bubbles grow at common interest rate:

$$P_{it} = V_{it} + B_{it} = V_{it} + \theta_i B_t$$

- Allow fundamental differences rents miss: $V_{it} = \kappa_i \psi_i V_t$
 - land quality: values cover all grades, rents only corn and soybean land
 - property taxes, risk, expected rent growth
- Normalized land price:

$$X_{it} := P_{it}/\psi_i = \kappa_i V_t + \gamma_i B_t, \quad \gamma_i := \theta_i/\psi_i$$

- Two testable implications
 - data can reject no-bubble benchmark
 - rejection does not by itself identify bubble

Implication 1: relative prices

- No bubble: $\log X_{it} = \log \kappa_i + \log V_t$
 - relative price $\log X_{it} - \overline{\log X_t}$ constant
 - cross-district dispersion constant
- Gordon model with falling discount rate ρ :

$$P_{it}/r_{it} = \frac{1}{\rho - g_i}$$

high ratios rise proportionally more $\implies$ dispersion **rises**

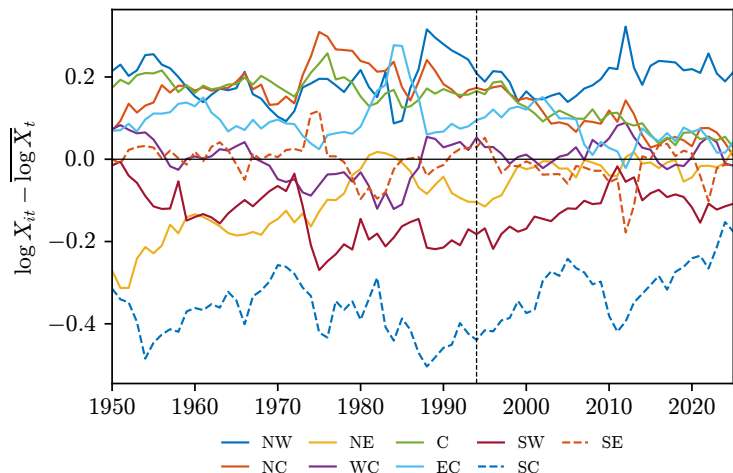
- Bubble common per unit of rent ($\gamma_i = \gamma$):

$$X_{it}/V_t = \kappa_i + \gamma B_t/V_t$$

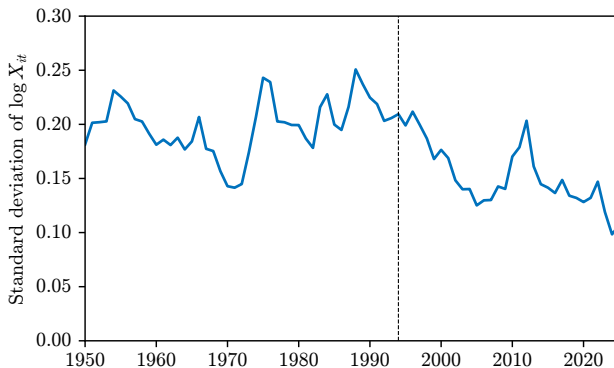
$\implies$ dispersion **falls** as B_t/V_t rises

- ψ_i only shifts each path by constant: test uses land values from 1950

Relative normalized land prices



Dispersion



- 1950–1993: 0.14–0.25, no trend
- 1994–2025: 0.21 $\rightarrow$ 0.11, trend -0.0023 (s.e. 0.0005)
- Gordon with fixed yield gaps: about $5\times$ actual in 2025

Implication 2: decomposition

- Base-period pattern: $\hat{\kappa}_i :=$ average of $X_{it}/\bar{X}_t$ over 1989–1993
- Substituting into $X_{it} = \kappa_i V_t + \gamma_i B_t$ yields cross-district regression

$$X_{it} = \tilde{V}_t \hat{\kappa}_i + \tilde{B}_t + u_{it}$$

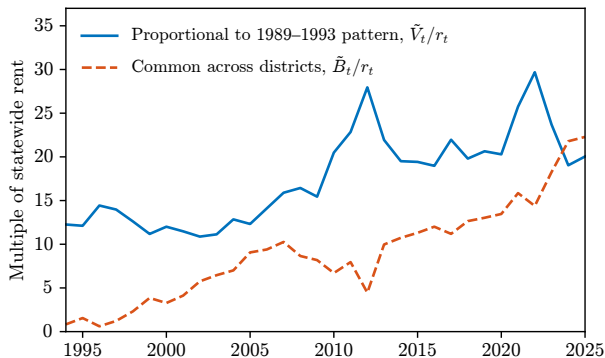
where

$$\tilde{V}_t = \bar{X}_0 V_t / V_0,$$

$$\tilde{B}_t = \bar{\gamma} V_t (B_t / V_t - B_0 / V_0)$$

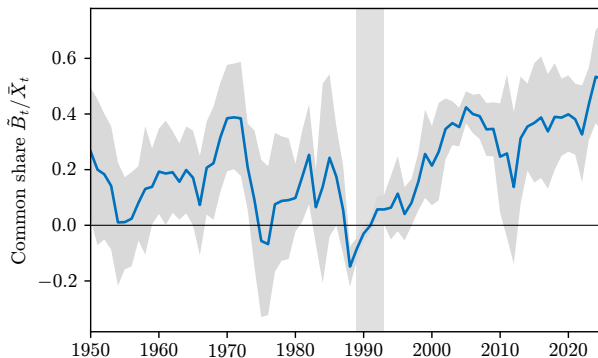
- No bubble: $\tilde{B}_t = 0$
- $\tilde{B}_t / \bar{X}_t$: share common across districts, growth of bubble relative to fundamental value since base period
- Requires γ_i uncorrelated with $\hat{\kappa}_i$

Decomposition in units of rent



- Average normalized price–rent ratio: 13.1 (1994) $\rightarrow$ 42.3 (2025)
- Common component contributes 21.5 of increase of 29.2

Common share



- 6% (1994) → 53% (2025); increase from 1994–1999 to 2020–2025: 0.32 [0.07, 0.56]
- Wild bootstrap band (9 districts)

Interpretation

- Before late 1990s: common share moves up and down, each rise reverses
 - 1970s boom: share falls ($0.39 \rightarrow -0.07$), dispersion rises
 - consistent with fundamental boom (Just and Miranowski, 1993)
 - early 1980s rise reverses in farm crisis
- Late 1990s onward: 0.08 (1997) $\rightarrow 0.42$ (2005) $\rightarrow 0.53$ (2024–2025)
 - coincides with dot-com bubble and US housing boom
 - unlike earlier rises, has not reversed
- Consistent with—but not proof of—land bubble
 - stable relative fundamentals $\kappa_i \psi_i$
 - bubble exposure uncorrelated with base-period pattern

Conclusion

- Debt rollover possible if and only if land price contains bubble
- Land share bounded away from zero: no bubble, efficiency, no rollover
- Vanishing land share and low autarky interest rate: bubble necessary, rollover possible, efficiency
- Rollover need not reflect dynamic inefficiency
- Iowa farmland: declining cash yield and rising common component since late 1990s
- Extensions: aggregate uncertainty, housing

Proof of Theorem: finite value of output

- $\liminf_{t \rightarrow \infty} s_t^X > 0$: there exist $\delta > 0$ and T such that

$$r_t \geq \delta Y_t \quad \text{for all } t \geq T$$

- Land price finite and $P_0 \geq V_0$:

$$\sum_{t=1}^{\infty} q_t r_t = V_0 \leq P_0 < \infty$$

- Hence

$$\sum_{t=0}^{\infty} q_t Y_t \leq \sum_{t=0}^{T-1} q_t Y_t + \frac{1}{\delta} \sum_{t=T}^{\infty} q_t r_t < \infty$$

- Because $E_t = Y_t - r_t \leq Y_t$, aggregate non-asset income has finite value:

$$\sum_{t=0}^{\infty} q_t E_t < \infty$$

Proof of Theorem: fundamental land price

- Multiply sequential budget constraints by q_t and sum:

$$\sum_{t=0}^T q_t(c_{it} - e_{it}) + q_T P_T x_{iT} = (P_0 + r_0)x_{i,-1}$$

- Finite value of income and no-Ponzi: lifetime budget constraint

$$\sum_{t=0}^{\infty} q_t c_{it} \leq \sum_{t=0}^{\infty} q_t e_{it} + (P_0 + r_0)x_{i,-1},$$

binding by local nonsatiation

- Sum across agents and use market clearing $\sum_i c_{it} = E_t + r_t$:

$$\sum_{t=0}^{\infty} q_t(E_t + r_t) = \sum_{t=0}^{\infty} q_t E_t + P_0 + r_0 \implies P_0 = \sum_{t=1}^{\infty} q_t r_t = V_0$$

Proof of Theorem: Pareto efficiency

- Suppose feasible (c'_i) Pareto dominates equilibrium (c_i)
- Optimality and local nonsatiation:

$$\sum_{t=0}^{\infty} q_t c'_{it} \geq \sum_{t=0}^{\infty} q_t c_{it},$$

strict for at least one agent

- Summing (all values finite):

$$\sum_{t=0}^{\infty} q_t \sum_i c'_{it} > \sum_{t=0}^{\infty} q_t (E_t + r_t),$$

contradicting feasibility

- Standard first welfare theorem; finite value of resources is key
- Adapts Santos and Woodford (1997); no restriction on demographics or preferences

Bubble necessity: proof idea

- Suppose $\liminf_{t \rightarrow \infty} P_t/a_t = 0$
 - along subsequence, asset trade becomes negligible
 - normalized consumption approaches autarky $(1, G\omega)$
 - interest rate approaches $R^a = f(1, G\omega)$
- But rents grow at $G_r > R^a$: present value of rents explodes, contradicting $P_t \geq V_t$
- Hence $P_t \geq ca_t$ for some $c > 0$, and

$$\frac{r_t}{P_t} \leq \frac{1}{c} \frac{r_t}{a_t},$$

which decays geometrically because $G_r < G$

- Bubble characterization: $\sum_t r_t/P_t < \infty \implies$ bubble

[◀ Return](#)[▶ Bubble characterization](#)

Characterizing bubbles

- Iterating no-arbitrage condition:

$$q_t P_t = \sum_{s=t+1}^T q_s r_s + q_T P_T$$

- Letting $T \rightarrow \infty$:

$$P_t = V_t + \frac{1}{q_t} \lim_{T \rightarrow \infty} q_T P_T$$

- Land price fundamental if and only if **no-bubble condition** holds:

$$\lim_{T \rightarrow \infty} q_T P_T = 0$$

Lemma (Bubble characterization, Montrucchio, 2004)

Land price contains bubble if and only if $\sum_{t=0}^{\infty} r_t / P_t < \infty$.

Proof of debt–bubble equivalence

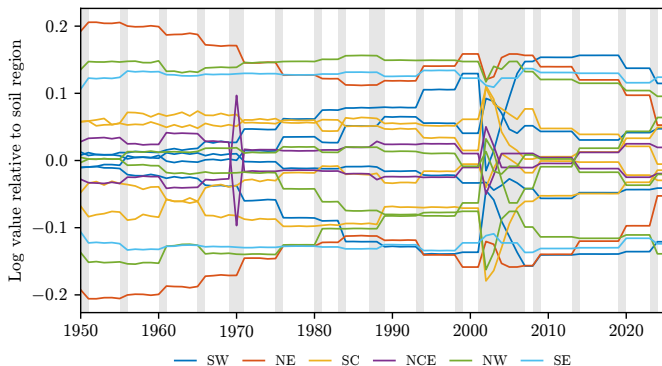
- ($\Rightarrow$) Debt equilibrium with land price $\tilde{P}_t$, holdings (x_{it}, d_{it})
 - consolidate: $P_t := \tilde{P}_t + D_t$, $z_{it} := (\tilde{P}_t x_{it} + d_{it})/P_t$
 - cost and payoff unchanged:

$$P_t z_{it} = \tilde{P}_t x_{it} + d_{it}, \quad (P_{t+1} + r_{t+1}) z_{it} = R_t (\tilde{P}_t x_{it} + d_{it})$$

- date-0 transfer $D_0 x_{i,-1} =$ capital gain on initial land
 - bubble $B_t = (\tilde{P}_t - V_t) + D_t \geq D_t > 0$
- ($\Leftarrow$) Bubbly equilibrium without debt
 - set $D_t := \theta B_t$ with $\theta = D_0/B_0 \in (0, 1]$, $\tilde{P}_t := P_t - D_t$
 - $B_{t+1} = R_t B_t$ implies $D_{t+1} = R_t D_t$
 - hold $x_{it} = z_{it}$ and $d_{it} = D_t z_{it}$

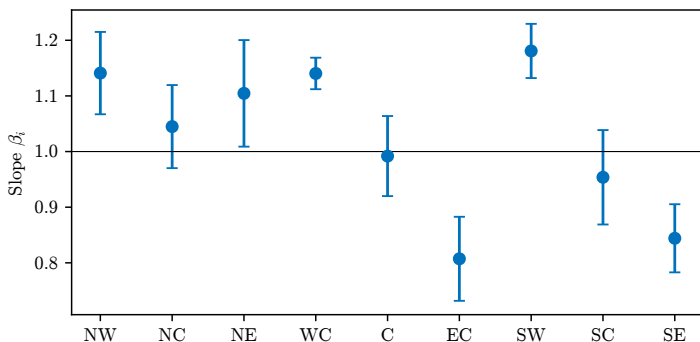
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County values move together by construction



- Interior counties relative to soil region; shaded: census-update years
- Soil region $\times$ year effects explain 97.4% (all counties) and 99.99% (interior counties) of annual changes

District rent slopes



- $\log r_{it} = \alpha_i + \beta_i \log r_t + \epsilon_{it}$
- Wald test of equal slopes: $\chi^2(8) = 531$ (Newey–West)
- Relative rents drift slowly

Wild bootstrap

- Only 9 districts: resampling districts performs poorly
- For each year, residuals $\hat{u}_{it}$, rescaled by leverage (HC2)
- Draw $X_{it}^* = \hat{V}_t \hat{\kappa}_i + \hat{B}_t + v_i \tilde{u}_{it}$
 - v_i : six-point distribution (Webb, [2023](#))
 - same v_i for district in all years
- 19,999 draws, conditional on $\hat{\psi}_i$ and $\hat{\kappa}_i$
- Assumes deviations independent across districts

[Return](#)

Robustness

- Measures using district rents (base 1994–1999): common share in 2025
 - fitted rents $P_{it}/\hat{r}_{it}$: 0.50
 - observed rents P_{it}/r_{it} : 0.45
- Base periods 1979–1983, 1985–1989, 1994–1999: 0.54, 0.53, 0.51
- Dropping any single district: 0.49–0.66; dispersion trend significant
- Acreage weighting: cash yield almost unchanged

[◀ Return](#)

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